

Time & Work – APTITUDE LEARNING NOTES

Complete concepts • formulas • shortcuts • solved examples • practice

1. Basic Work Concept

If a person completes work in D days, one-day work = $1/D$.

FORMULA: Daily rate = $1/\text{time}$; Time = work/rate .

When people work together, add their daily work rates.

2. LCM Work Method

Take total work as the LCM of individual times. Then daily work becomes an integer.

Example: 12-day and 18-day workers → total work 36 units; rates 3 and 2 units/day.

3. Efficiency and Time

For the same work, efficiency is inversely proportional to time.

FORMULA: Efficiency ratio = reciprocal of time ratio.

If A is 50% more efficient than B, efficiency A:B=3:2 and time A:B=2:3.

4. Men, Days and Hours

For equal worker efficiency and same work:

FORMULA: $M_1 D_1 H_1 = M_2 D_2 H_2$.

Include efficiency ratios when workers are not equally efficient.

5. Alternate Work

Add rates for simultaneous work. For alternate days, calculate work completed in each cycle, then handle the remaining fraction.

Check carefully who works on the final day.

6. Pipes and Cisterns

Filling rate is positive; emptying/leak rate is negative.

FORMULA: Net rate = sum of filling rates - sum of emptying rates.

Time = $1/\text{net rate}$ when capacity is treated as 1.

7. Wages

Wages are proportional to work done.

FORMULA: Wage ratio = work ratio = efficiency × time ratio.

8. Two-Person Shortcut

If A takes a days and A+B take b days, B's rate = $1/b - 1/a$.

FORMULA: B's time = $ab/(a-b)$, when $a > b$.

Solved Aptitude Examples

Q1. A takes 12 days and B 18 days. Together?

Solution: $\text{Rate} = 1/12 + 1/18 = 5/36 \rightarrow 36/5 = 7.2 \text{ days}$.

Q2. A takes 20 days, B 30 days. Together for 6 days; fraction remaining?

Solution: Combined rate $= 1/12$; 6 days complete $1/2 \rightarrow 1/2 \text{ remains}$.

Q3. A alone 15 days; A+B 10 days. B alone?

Solution: $1/10 - 1/15 = 1/30 \rightarrow 30 \text{ days}$.

Q4. 12 workers finish in 18 days. Workers for 12 days?

Solution: $12 \times 18 = M \times 12 \rightarrow 18$.

Q5. A twice as efficient as B; B takes 24 days.

Solution: A takes **12 days**.

Q6. Pipes fill in 6 h and 8 h. Together?

Solution: $1/6 + 1/8 = 7/24 \rightarrow 24/7 \text{ h}$.

Q7. Fill pipe 10 h, leak 15 h. Net time?

Solution: $1/10 - 1/15 = 1/30 \rightarrow 30 \text{ h}$.

Q8. A,B,C take 10,15,30 days. Together?

Solution: Rates sum $= 1/5 \rightarrow 5 \text{ days}$.

Q9. A works 4 days on a 12-day job. Fraction left?

Solution: Completed $1/3 \rightarrow 2/3 \text{ remains}$.

Q10. 8 men work 6 h/day for 15 days. Equivalent men at 10 h/day for 12 days?

Solution: $8 \times 6 \times 15 = M \times 10 \times 12 \rightarrow 6 \text{ men}$.

Practice Questions

1. A does a job in 16 days and B in 24. Together time?

2. A takes 18 days, B 36. Together for 6 days; fraction remaining?

3. A alone 12 days; A+B 8 days. B alone?

4. 15 workers finish in 20 days. Workers for 12 days?

5. A is 3 times as efficient as B. B takes 30 days. A's time?

6. Pipes fill in 12 h and 18 h. Together?

7. Pipe fills in 8 h; leak empties in 24 h. Net time?

8. A,B,C take 12,18,36 days. Together?

9. A completes $1/5$ of work in 3 days. Whole work time?

10. 10 men work 8 h/day for 9 days. Equivalent workers for 12 h/day for 6 days?

11. A+B take 6 days; A alone 10 days. B alone?

12. A takes 10 days, B 15 days. If they alternate starting A for 10 days, fraction completed?

13. A is 25% more efficient than B. B takes 20 days. A's time?

14. 4 workers complete a job in 30 days. Total work in worker-days?

15. Pipe fills in 5 h, B in 10 h, leak empties in 20 h. Net time?

Answer Key

1) $48/5=9.6$ days • 2) $1/2$ • 3) 24 days • 4) 25 workers • 5) 10 days • 6) $36/5$ h • 7) 12 h • 8) 6 days • 9) 15 days • 10) 10 workers • 11) 15 days • 12) $2/3$ • 13) 16 days • 14) 120 worker-days • 15) $20/3$ h

Quick Revision Formula Sheet

- Daily work= $1/\text{time}$
- Together: rates add
- A+B: $1/T=1/A+1/B$
- Men $\times$ days $\times$ hours constant for equal efficiency
- Efficiency $\propto 1/\text{time}$
- Pipes: filling positive, leak negative
- B time= $ab/(a-b)$ when A=a and A+B=b

Exam strategy: Identify the pattern first, choose the shortest valid formula, solve step-by-step, and verify the final answer.